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SAR Satellites including the NASA-ISRO SAR Mission

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with NISAR slides from

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Classic SAR satellites



Satellite	repeat cycle (days)	wave-length (cm)
European ERS-1/ERS-2 '92-'01(-2011)	35 (1,3,183)	6
Canadian Radarsat-1 1995-2013	24	6
European Envisat '03-Sep.'10('10-Apr.'12)	35 (30)	6
Japanese ALOS Jan. 2006–Apr. 2011	46	24
German TerraSAR-X '07, TanDEM-X '10	11	3
Italian COSMO-SkyMed 4x launched '07-'10	16 (1,4,7,8)	3
Canadian Radarsat-2 launched Dec. 2007	24	6

SAR satellites processing

satellite	Modes	ISCE2 processing support?
ERS-1/ERS-2	1 strip map	yes raw, SLC
RADARSAT-1	7 standard strip 5 fine strip ScanSAR	strip: maybe CEOS raw ScanSAR: no
Envisat	7 standard strip (IM) ScanSAR (WS)	strip: yes raw, SLC ScanSAR: no
ALOS	strip FBS, FBD, POL ScanSAR	strip: yes raw, SLC ScanSAR: no
TerraSAR-X, PAZ	strip ScanSAR Spotlight	strip: SLC ScanSAR: no Spotlight: SLC
COSMO/SkyMed	strip, ScanSAR, Spotlight	strip: raw, SLC Spotlight: SLC
RADARSAT-2	strip (4 types), ScanSAR, Spotlight	strip: SLC Spotlight: ?

other SAR spacecraft

Satellite	repeat cycle (days)	wave- length (cm)
US SeaSat 1978 (110 days)	17/3	24
Japanese JERS-1 1992-1998	44	24
US Shuttle Imaging Radar-C (SIR-C) 1994 (2x 10-day flights)	1 day, 6 months	3, 6, 24



new SAR spacecraft



satellite (launch)	repeat cycle (days)	wave- length (cm)
Copernicus Sentinel-1 (A: Apr. 2014–June 2026, B: Apr. 2016–Dec. 2021, C: Dec. 2024, D: Nov. 2025)	12(6)	6
Japanese ALOS-2 (May 2014)	14	24
Indian RISAT-1 (Apr. 2012–Sept. 2016)	25	6
NASA-ISRO SAR (NISAR) mission (July 2025)	12	12,24



New SAR satellites processing



Satellite	Modes	ISCE2 proc.	ISCE3 proc.
Sentinel-1A, B, C, D	stripmap TOPS	strip: SLC (raw extra work) TOPS: SLC (topsApp.py)	TOPS: SLC (COMPASS)
ALOS-2 (alos2App.py, alosStack)	strip (3 types) ScanSAR Spotlight	strip: SLC ScanSAR: full-aperture SLC Spotlight: SLC	Strip: SLC (testing)
RISAT-1	stripmap ScanSAR	strip: SLC, raw extra work ScanSAR: no	

More new SAR spacecraft

satellite (launch or planned)	repeat cycle (days)	wave-length (cm)
Argentina SAOCOM-1 (A: Oct. 2018, B: Aug. 2020)	16(8)	24
Japanese ALOS-4 (June 2024)	14	24
Italian COSMO-SkyMed 2nd Gen. (1: Dec. 2019, 2: Jan. 2022, 3:, 4:)	N/A	3
Canadian RADARSAT Constellation Mission 3x (Jun. 2019)	4	6



More new SAR spacecraft



Satellite (launch or planned)	Repeat cycle (days)	Wavelength (cm)
South Korea KompSat-5 (2013-2022)	16(8)	3
Spanish PAZ (clone TerraSAR-X) (Feb. 2018)	11 (4,7)	3
Capella-1 through -n (2018–present) some InSAR-capable	?	3
ICEYE ~40 operational satellites now (in July 2026) some satellites have 1-day separation	1	3
Umbra 24 satellites planned (first launched 2021), no InSAR yet	N/A	3
Several other countries and companies have launched SAR and InSAR-capable satellites		

It is time to get ready



- NISAR launched July 30, 2025!!
- First light images ~2 months after launch
- Science mode active ~3 months after launch
- Full science operations December 2025
- Global products to Level 2 are fully and openly available to the global community since July 2026
- Broad scientific and applied uses
- Cloud-based data, tools and services facilitate access and use
- Data processed and stored in Amazon cloud (AWS)

For more information: <https://science.nasa.gov/mission/nisar/>
News updates: <https://science.nasa.gov/blogs/nisar/>



Open-Source Science for NISAR



- Science products
 - ASF Vertex and EarthData searchable
 - NISAR HDF5 specification: <https://science.nasa.gov/mission/nisar/data/>
- Open Source Software – SDS and data processing code available for download
 - InSAR Scientific Computing Environment, Enhanced Edition (ISCE3): <https://github.com/isce-framework/isce3>
- Open Source Science algorithms for science products
 - Jupyter notebooks available for download: <https://gitlab.com/nisar-science-algorithms>
- Open Source Training Opportunities
 - Jupyter notebooks in cloud training environments at Alaska Satellite Facility OpenScienceLab
 - ARSET and other courses: <https://science.nasa.gov/mission/nisar/sar-education-resources/>
 - Cloud computing resources for NASA subscribers